

# GeminiChatProvider.java

```
package com.harisudhan.harisudhanmart.ai;

import com.google.gson.JsonObject;
import com.google.gson.JsonParser;
import java.io.IOException;
import java.util.concurrent.TimeUnit;
import okhttp3.MediaType;
import okhttp3.OkHttpClient;
import okhttp3.Request;
import okhttp3.RequestBody;
import okhttp3.Response;
import org.slf4j.Logger;
import org.slf4j.LoggerFactory;

/**
 * Real provider implementation – calls an LLM API server-side. The API key is read from an
 * environment variable / config.properties and never sent to the client (Section 11, rule 1).
 * Wrapped in try/catch by ChatService; on failure the service falls back to MockChatProvider.
 */
public class GeminiChatProvider implements ChatProvider {

    private static final Logger LOG = LoggerFactory.getLogger(GeminiChatProvider.class);
    private static final String ENDPOINT =
        "https://generativelanguage.googleapis.com/v1beta/models/gemini-1.5-flash:generateContent";
    private static final String PROMPT_TEMPLATE =
        "You are the support assistant for an e-commerce marketplace called HarisudhanMart. "
        + "Only answer questions about products, orders, shipping, payments, and reviews. "
        + "Keep answers under 3 sentences. Context: %s. Question: %s";

    private final OkHttpClient client;
    private final String apiKey;

    public GeminiChatProvider(String apiKey) {
        this.apiKey = apiKey;
        this.client = new OkHttpClient.Builder()
            .connectTimeout(5, TimeUnit.SECONDS)
            .readTimeout(8, TimeUnit.SECONDS)
            .build();
    }

    @Override
    public String getReply(String userMessage, String context) {
        try {
            String prompt = String.format(PROMPT_TEMPLATE, context == null ? "" : context, userMessage);
            JsonObject part = new JsonObject();
            part.addProperty("text", prompt);
            JsonObject content = new JsonObject();
            content.add("parts", com.google.gson.JsonParser.parseString(
                "[" + part.toString() + "]"));
            JsonObject body = new JsonObject();
            body.add("contents", com.google.gson.JsonParser.parseString(
                "[" + content.toString() + "]"));

            RequestBody requestBody = RequestBody.create(
                body.toString(), MediaType.parse("application/json"));
            Request request = new Request.Builder()
                .url(ENDPOINT + "?key=" + apiKey)
                .post(requestBody)
                .build();

            try (Response response = client.newCall(request).execute()) {
                if (!response.isSuccessful() || response.body() == null) {
                    throw new IOException("Non-2xx response from chat provider: " + response.code());
                }
            }
        } catch (IOException e) {
            LOG.error("Error in GeminiChatProvider.getReply: ", e);
            return null;
        }
    }
}
```

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        }
        JsonObject json = JsonParser.parseString(response.body().string()).getAsJsonObject();
        return json.getAsJsonArray("candidates").get(0).getAsJsonObject()
            .getAsJsonObject("content").getAsJsonArray("parts").get(0)
            .getAsJsonObject().get("text").getString();
    }
} catch (Exception e) {
    LOG.error("GeminiChatProvider call failed", e);
    throw new RuntimeException(e);
}
}
```